SOLUTION

SECTION-I

Single Correct Answer Type **15 Q. [3 M (–1)]**

1. Ans. (C)

2. Ans. (B)

3. Ans. (D)

4. Ans. (B)

5. Ans. (A)

6. Ans. (B)

7. Ans. (C)

8. Ans. (B)

9. Ans. (B)

Sol. Equation of demped oscillation

$$x = \left[Ae^{-b/2m t} \right] \sin \omega t$$

$$I = \left[I_0 e^{-\frac{R}{2L}t} \right] \sin \omega t$$

$$I_0 e^{-(R/2L)t} = \frac{I_0}{e} \qquad \text{For maximum}$$

$$\frac{R}{2L}t = 1 \qquad \phi = \omega t$$

$$t = \frac{2L}{R} \qquad \phi = \frac{2L\omega}{R} \left(X_L = \omega L \right)$$

$$(2\pi) n = \frac{2 \times 200}{20} = n = \frac{10}{\pi}$$

10. Ans. (D)

Sol. $\omega' = \sqrt{\frac{K}{m} - \frac{b^2}{4m^2}}$ and $\omega' = \frac{2\pi}{T}$

11. Ans. (D)

Sol. This is the case of Damped oscillation.

12. Ans. (C)

Sol. Considering gravity, buoyant force and viscouse force oscillation will be damped.

$\Omega_1^2 = \omega_2^2 - C$, ω_2 is natural frequency.

$\Rightarrow T_1 > T_2$ C is a constant.

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13. Ans. (B)

14. Ans. (C)

15. Ans. (B)

Sol. $T_0 = 2\pi\sqrt{\frac{m}{k}}$

$= \frac{2\pi}{\sqrt{10}}$

$A = A_0 e^{-t/\gamma}$

$\therefore$ for $A = \frac{A_0}{e}$, $t = \gamma$

$t = \gamma = \frac{2m}{b} = \frac{2m}{\frac{B^2\ell^2}{R}} = 10^4 s$

$\therefore$ No of oscillation $\frac{t}{T_0} = \frac{10^4}{2\pi/\sqrt{10}} \approx 5000$.

Multiple Correct Answer Type **2 Q. [4 M (–1)]**

16. Ans. (A,B,D)

17. Ans. (A,D)

Linked Comprehension Type **(1 Para × 2Q.) [3 M (-1)]**
(Single Correct Answer Type)

18. Ans. (A)

19. Ans. (D)

SECTION-II

Linked Comprehension Type **(1 Para × 2Q.) [3 M (-1)]**
(upto second decimal place)

20. Ans. 50.00

21. Ans. 30.00

Numerical Answer Type Question **1Q.[3(0)]**
(upto second decimal place)

22. Ans. 0.4

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